



Agglomerative Clustering

Agglomerative clustering is a hierarchical method for grouping data points based on their similarities.

Linkage Overview



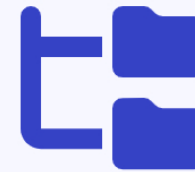
Linkage Definition

Linkage indicates the method used to calculate the distance between groups of observations to form clusters.



Importance of Linkage

Different linkage methods can lead to significantly different clustering results, impacting analysis outcomes.



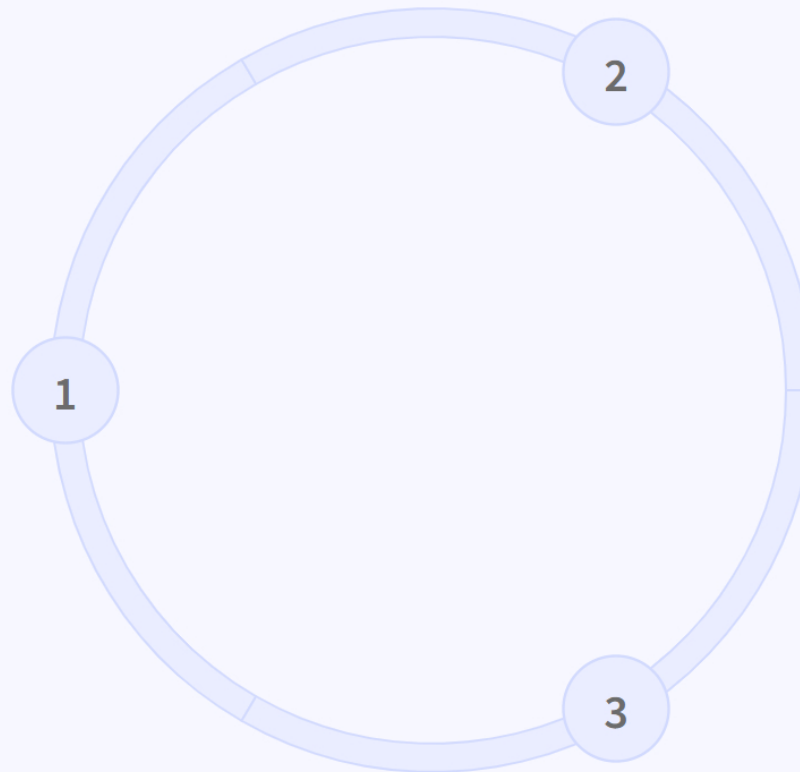
Hierarchical Structure

Agglomerative clustering builds a hierarchy of clusters, which provides a visual representation of the data through dendrograms.

Single Linkage Clustering

Min Distance

Merges clusters based on the minimum distance between any two points in different clusters.



High Noise Sensitivity

Highly sensitive to noise and outliers, potentially leading to incorrect cluster formation.

Best Use Case

Ideal for datasets focusing on connectivity with long, chain-like clusters.



Complete Linkage Clustering

Max Distance

Calculates using maximum distance between points in clusters, favoring compact clusters.

Medium Noise Sensitivity

Less sensitive to noise than single linkage, but still affected by outliers.

Best Use Case

Useful for identifying clusters in tight or well-separated datasets.

Average Linkage

1

Mean Distance

Average linkage uses the average distance between points in different clusters.

2

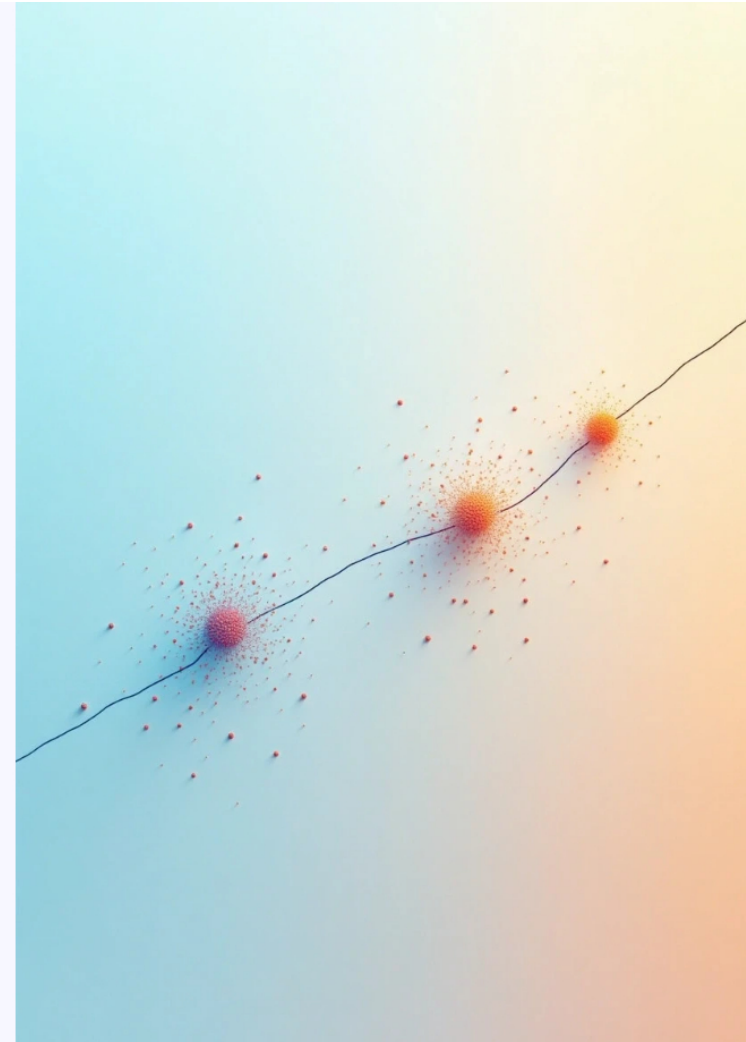
Medium Noise Sensitivity

It strikes a balance between sensitivity to noise and meaningful clusters identification.

3

Best Use Case

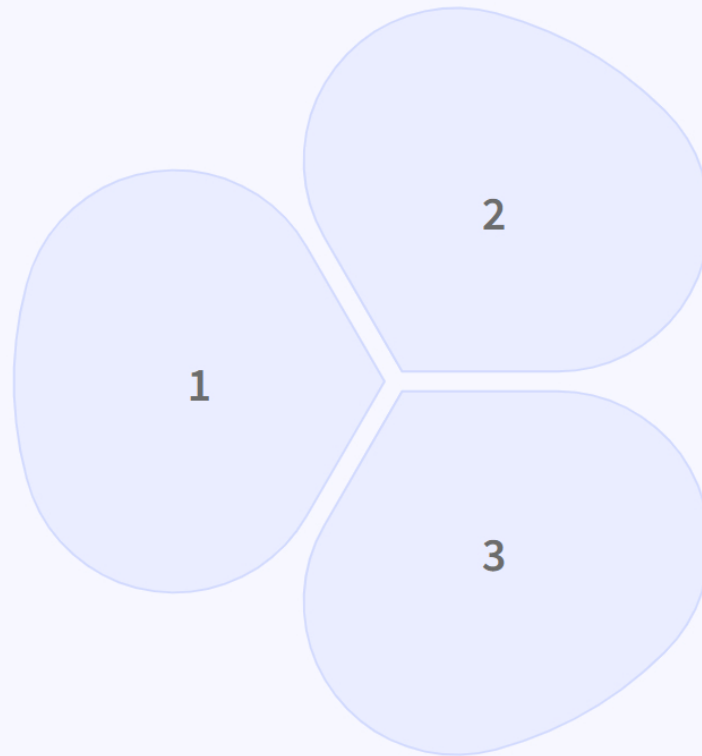
This method is suitable for **General clustering** tasks with no specific shape anticipated.



Ward's Linkage

Variance Approach

Minimizes variance within clusters for better purity.



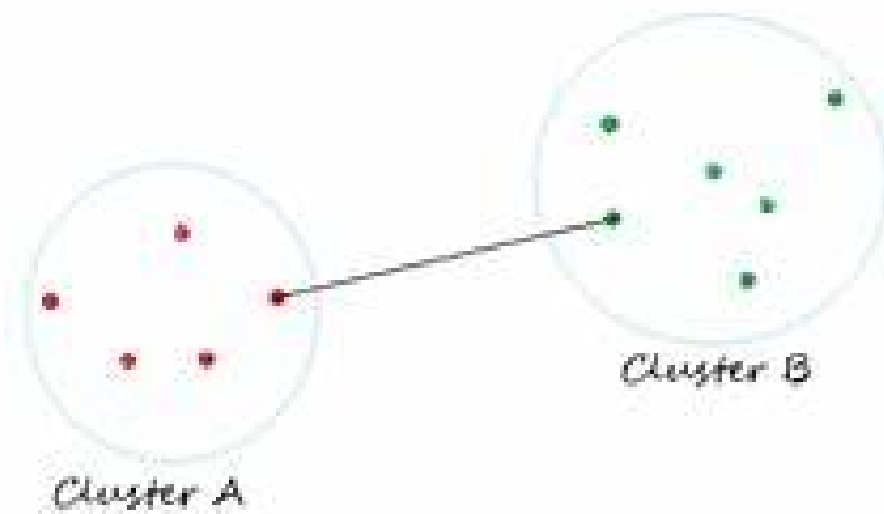
Low Noise Sensitivity

Resistant to noise, ensuring robust clustering.

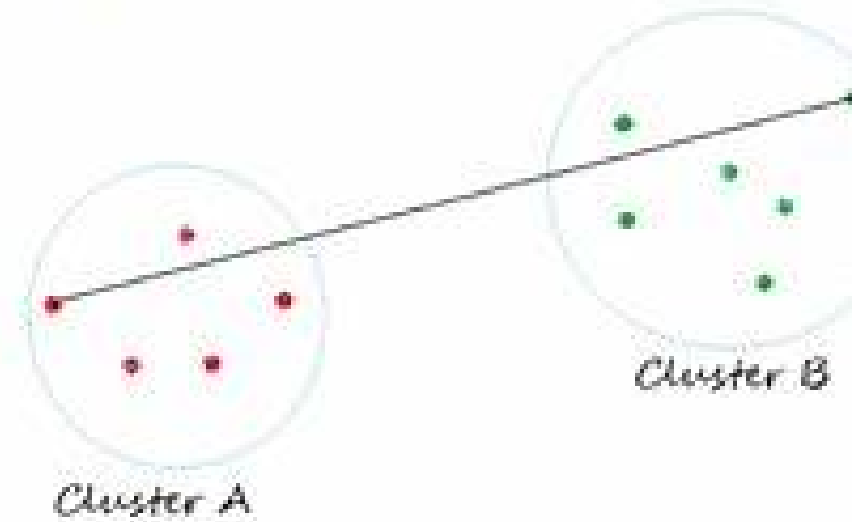
Best Use Case

Ideal for **clean segmentation** in markets and images.

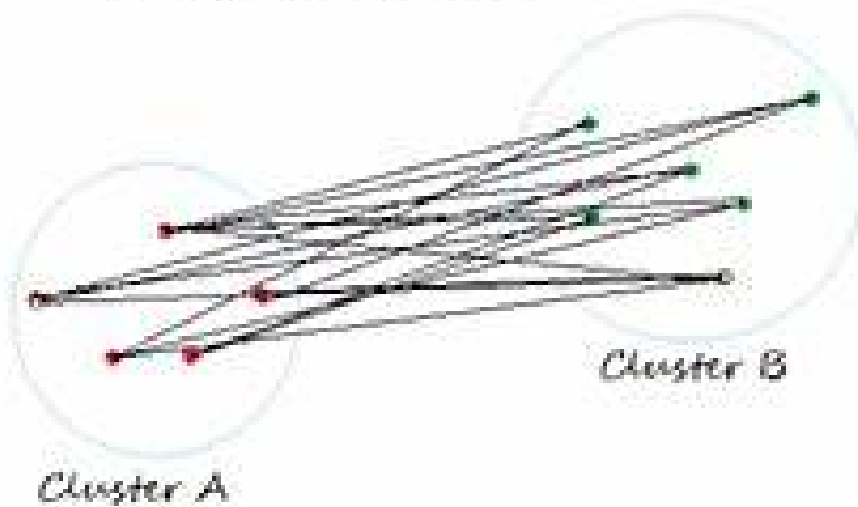
Single Linkage



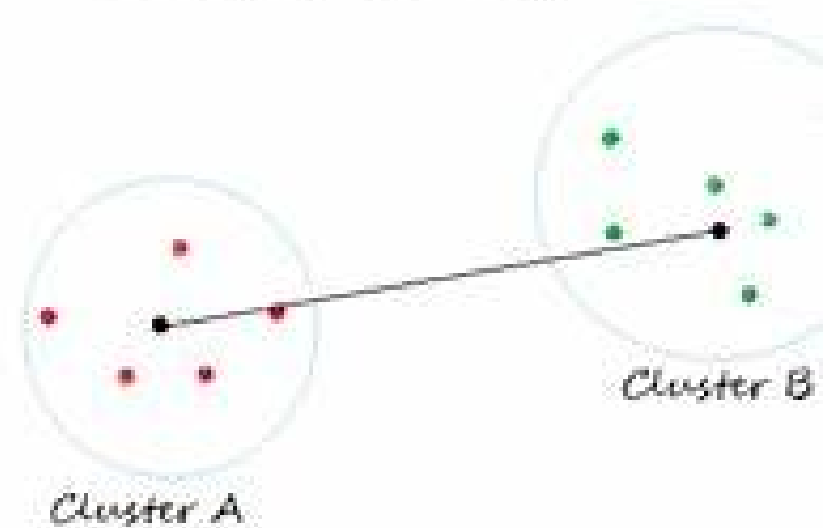
Complete Linkage



Average Linkage



Centroid Linkage



Quick Comparison of Linkage Types

Linkage Type	Distance Based On	Shape	Noise Sensitivity	Best Use Case
Single	Min distance	Arbitrary	High	Connectivity
Complete	Max distance	Compact	Medium	Tight clusters
Average	Mean distance	Balanced	Medium	General use
Ward	Variance	Spherical	Low	Clean segmentation



Conclusion on Cluster Analysis

Choosing Linkage Method

The choice of linkage method greatly influences the resulting clusters and their interpretation.

Cluster Evaluation

Post-clustering evaluations can help validate the chosen method.

Versatile Applications

Agglomerative clustering is applied across various fields for pattern recognition.